Variable-rate streaming with a summed-spiking readout

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Abstract

This experiment will test whether PING networks trained across input rates can classify continuous MNIST streams without the fixed-rate and output-membrane mismatch inherited by exp048. The trained readout sums excitatory spikes, and inference uses exactly the same operation over a window equal to the current digit's presentation duration. The planned results comprise matched-condition streaming, variable-rate and variable-duration streaming, a 200 ms rate psychometric, and a duration-by-rate accuracy map. Results remain pending until exp022 produces all three variable-rate checkpoints.

Methods

1. Training source

The experiment loads `ping__variable_rate__seed42`, `ping__variable_rate__seed43`, and `ping__variable_rate__seed44` from exp022. Each PING network is trained by sampling one maximum-pixel Poisson rate independently per image presentation from 0.5, 1, 2, 5, 10, and 25 Hz. The recurrent weights and classifier are frozen throughout this experiment.

Note. The exp022 cells are registered and `tools/snn` implements their variable-rate training interface, but the full Cambridge jobs have not run yet. This entry remains their downstream target and fails loudly rather than falling back to fixed-rate weights.

2. Summed-spiking readout

For a presentation beginning at timestep a and ending before timestep b, class evidence is

$$\boldsymbol{z} = \left(\sum_{t=a}^{b-1} \boldsymbol{s}^{E(t)} \right) W_{\text{out}}. \quad (1)$$

Here $\boldsymbol{s}^{E(t)}$ is the 1024-element excitatory spike vector at timestep t, W_{out} is the trained 1024-by-10 readout matrix, and $\boldsymbol{z}$ contains the ten class logits. The predicted digit is the index of the largest logit. This is the trained `rate` readout: no output LIF membrane, leak constant, or post-hoc `mem-mean` reconstruction is introduced.

3. Matched inference

Every digit is read from only the spikes generated during its own presentation:

$$T_{\text{readout}} = T_{\text{presentation}}. \quad (2)$$

Here T_{readout} is the spike-summation window and $T_{\text{presentation}}$ is the duration of the corresponding digit. The first protocol holds both at 200 ms and presents a five-digit stream. This checks the trained duration and readout before either input rate or duration is varied.

4. Variable-rate and variable-duration inference

A second five-digit stream changes both presentation duration and maximum-pixel Poisson rate at digit boundaries. Each digit retains its own matched readout window. A factorial evaluation then crosses presentation durations 25, 50, 100, and 200 ms with the six training rates. Every cell is evaluated independently for seeds 42, 43, and 44.

5. Psychometric protocol

The rate psychometric fixes presentation and readout duration at 200 ms, varies the maximum-pixel rate across the six training values, and reports held-out classification accuracy across the three trained seeds. This isolates rate robustness from shortened evidence windows.

Results

1. Matched-condition stream

TODO. Reproduce the structure of exp048′s streaming figure: excitatory and inhibitory rasters, digit boundaries, and the ten online class-evidence traces. Unlike exp048, every trace will be formed by accumulating spikes from the start of the current digit.

2. Variable conditions

TODO. Reproduce exp048′s joint temporal and encoding-rate summary using the variable-rate-trained weights. Pair the duration-by-rate accuracy map with the 200 ms psychometric curve, so the effects of rate and available integration time remain distinguishable.

3. Comparison with exp048

TODO. Compare the completed results with exp048. The comparison must be interpretive rather than a silent replacement. Exp048 uses fixed-25-Hz training and reconstructs a sliding `mem-mean` output. This experiment uses mixed-rate training and the trained summed-spiking readout. Any accuracy difference therefore reflects the combined training-distribution and readout correction, not a single isolated intervention.